

Prachatos Mitra

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SUMMARY

Computer architecture PhD candidate in performance modeling and bottleneck analysis for large-scale systems, with performance analysis and hypervisor systems work experience at AMD and Nutanix respectively. Seeking Summer 2027 performance modeling / architecture internships.

EDUCATION

Georgia Institute of Technology

January 2023 - December 2027 (expected)

PhD, Computer Science

GPA: 4.0/4.0

Advisor: Dr. Alexandros Daglis

Research Area: Performance modeling and bottleneck analysis for large-scale systems; scale-out graph processing architectures

Indian Institute of Science

August 2018 - July 2020

M.Tech., Computer Science

GPA: 8.80/10.00

M.Tech thesis advisor: Dr. Arkaprava Basu

Research Area: Memory systems - TLB dead page and cache dead-block prediction

Maulana Abul Kalam Azad University of Technology

August 2014 - July 2018

B.Tech., Computer Science and Engineering

GPA: 9.01/10.00

RESEARCH EXPERIENCE

• Optimizing resource allocation for actor-based workloads on commodity hardware

Georgia Institute of Technology

2026-Present

- Profiled performance characteristics of actor-based workloads on commodity hardware as a function of CPU frequency, LLC size and memory bandwidth
- Designing a runtime-level detector for imbalance of resources and work and dynamically adjusting resources

• Design and simulation for actor-model based sparse graph processing architecture

Georgia Institute of Technology

2023-Present

- Helped design messaging and network-connected components for a novel scale-out actor-model graph processing architecture, under the IARPA AGILE program
- **MacroPattern:** abstracted actor-model workloads into six fitted distributions per handler, scaled analytically to arbitrary problem and system size via two-level curve fitting
- **MPSim:** SST-based parallel discrete-event simulator consuming scaled distributions; simulates ~16 million processing elements at ~8 MB per simulated core. 2.5–30× more accurate than regression-based scaling across graph microbenchmarks; workload representation seven orders of magnitude smaller than equivalent traces
- Exposed a head-of-line blocking pathology in the target architecture's node-level queueing, visible only at scale

• Dead Page and Dead Block Predictors: Cleaning TLBs and Caches Together

Indian Institute of Science (Master's thesis)

2019-2020

- Demonstrated the presence of “dead” or unused entries in the L2 TLB
- Designed a dead-page predictor to improve TLB efficiency by bypassing such entries
- Designed a dead-block predictor for last level caches using TLB information
- Showed performance gains of over 8% with low storage overheads

WORK EXPERIENCE

AMD

Bangalore, India

Senior Silicon Design Engineer (Server Performance Group) September 2022 - December 2022

- Profiled networking and virtualization workloads on EPYC Bergamo servers using perf / hardware counters, characterizing IO path behavior
- Attributed throughput loss to OS interrupt handling overhead, amplified by Bergamo's core count
- Findings fed into research on utilizing user interrupts (uintr) as a potential kernel bypass path for interrupt delivery

Nutanix

Bangalore, India

Member of Technical Staff

August 2020 - September 2022

- Developed and maintained KVM/QEMU components of the Nutanix AHV hypervisor
- Built VMM live-upgrade support in Libvirt utilizing file-descriptor passing, presented at KVM Forum 2021
- Built a VMM migration monitoring feature, including developing a new daemon to monitor and analyze live migration status

PUBLICATIONS

- [1] **HPCA-27:** Chandrashis Mazumdar*, **Prachatos Mitra***, Arkaprava Basu, "Dead Page and Dead Block Predictors: Cleaning TLBs and Caches Together", Proceedings of 27th IEEE International Symposium on High-Performance Computer Architecture (HPCA-27) (virtual), February, 2021. Available at: <https://doi.org/10.1109/HPCA51647.2021.00050>
- [2] **SC-W '23:** **Prachatos Mitra**, Alexandros Daglis, "Filtering Wasteful Vertex Visits in Breadth-First Search", Proceedings of the SC '23 Workshops of The International Conference on High Performance Computing, Network, Storage, and Analysis, November 2023. Available at: <https://doi.org/10.1145/3624062.3625133>

TALKS AND PRESENTATIONS

- [1] **KVM Forum 2021:** **Prachatos Mitra** and Soham Ghosh "Support for Fast and Reliable VMM Live Upgrades in Libvirt", KVM Forum 2021 (virtual). Available at: https://www.youtube.com/watch?v=0C_0tcIJgpU

SERVICE

- [1] ASPLOS 2027: Artifact Evaluation Committee member

TECHNICAL SKILLS

Languages	C/C++, Python
Performance	perf, PAPI, hardware counters, roofline analysis
Simulation tools	SST, gem5, Sniper, PDES
Systems	Linux, Libvirt, QEMU, KVM, Bash, L ^A T _E X

*Equal contribution